WRITING SAMPLE · POLICY ANALYSIS & PROGRAM EVALUATION

Hotline Data Discrepancy Report

Identification and quantification of a duplication defect affecting a year of reported hotline volume, with the correction required to restore accurate counts.

DOCUMENT	**Data quality report**
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Introduction

This report addresses a critical data discrepancy identified in the source table that populates the agency's daily hotline decision summary report. The analysis covers reports received between 30 April 2023 and 30 April 2024.

Background and findings

While conducting an analysis of reports for the Hotline team, a discrepancy was found between the total number of reports shown in the reporting system's hotline report and the total pulled directly from the case management system. The two totals are set out in Table 1 and Table 2 below.

Table 1. Monthly hotline decision summary as reported by the agency reporting system, April 2023 – April 2024. Counts are aggregate; no case-level data is reproduced.

Period	Total reports	Assessment	Assessment %	Screen out	Screen out %	Information & referral	Service request
2023 — April	560	338	60.36%	128	22.86%	94	0
May	39,294	24,490	62.33%	12,012	30.57%	2,784	8
June	27,076	17,624	65.09%	7,218	26.66%	2,232	2
July	27,030	17,752	65.68%	7,010	25.93%	2,264	4
August	38,476	24,576	63.87%	11,238	29.21%	2,660	2
September	40,484	24,880	61.46%	12,560	31.02%	3,042	2
October	37,518	23,186	61.80%	11,550	30.79%	2,776	6
November	37,630	22,864	60.76%	12,104	32.17%	2,658	4
December	32,162	19,724	61.33%	9,746	30.30%	2,692	0
2024 — January	18,138	11,342	62.53%	5,492	30.28%	1,299	5
February	19,072	11,704	61.37%	6,074	31.85%	1,293	1
March	17,825	10,658	59.79%	5,776	32.40%	1,391	0
April	20,344	12,314	60.53%	6,678	32.83%	1,351	1
Total	**355,609**	**221,452**	**62.27%**	**107,586**	**30.25%**	**26,536**	**35**

Table 2. Reported totals compared against the source table and the case management system, on both a distinct and a non-distinct basis. System names have been generalized.

Outcome	Case management system (distinct, created date)	Source table (distinct, report date)	Reporting system report	Case management system (created date)	Source table (report date)
Information & referral	15,918	15,893	26,536	35,877	26,494
Screen in / assess	127,214	133,159	221,452	677,910	220,878
Screen out	71,401	65,475	107,586	360,008	107,268
Service request	21	21	35	66	35
Total	**214,554**	**214,548**	**355,609**	**1,073,861**	**354,675**

Based on the overall totals, the non-distinct source table report-date figure of **354,675** is the closest match to the reporting system total of **355,609**. The report-date column in the source table contains duplicates by download date, which indicates that the reporting system totals are affected by the same duplication.

Table 3 shows a sample of ten rows, representing five unique reports drawn from the source table. Each report identifier appears twice, with the download date the only column that differs across the rest of the record. This pattern is present across **140,114** distinct reports out of the 354,675 reports pulled from the source table. Those duplicated reports account for **280,228** of the 354,675 total records, when they should represent only 140,114. Consequently **39.5%** of the reports are duplicates, leaving only **60.5%** of the total as non-duplicated actual reports. On review, this duplication has been present since the period between 30 April 2022 and 30 April 2023.

Table 3. Sample of source-table records showing duplicated download dates. Report identifiers are surrogate values; the original identifiers have been replaced and the duplication pattern preserved.

Report identifier	Download date
RPT-000001	2024-12-30
RPT-000001	2024-12-31
RPT-000002	2024-12-31
RPT-000002	2024-12-30
RPT-000003	2024-12-31
RPT-000003	2024-12-30
RPT-000004	2024-12-31
RPT-000004	2024-12-30
RPT-000005	2024-12-31
RPT-000005	2024-12-30

Summary

- An analysis of reports for the Hotline team revealed a discrepancy between the total number of reports in the reporting system's hotline report and those pulled directly from the case management system.
- The daily hotline reports source table is used to generate the reporting system's daily hotline decision summary by referral report.
- The non-distinct source table report-date total of 354,675 is the closest match to the reporting system total of 355,609.
- The report-date column in the source table contains duplicates by download date, which indicates that the reporting system totals are affected by the same duplication.
- A sample of ten rows, representing five unique reports, showed each report identifier appearing twice with the download date as the only differing column.
- This pattern was observed across 140,114 distinct reports out of the 354,675 reports pulled from the source table.
- Those duplicated reports account for 280,228 of the 354,675 total records, when they should represent only 140,114.
- Consequently 39.5% of the reports are duplicates, leaving 60.5% as non-duplicated actual reports.